ADC Lab Logbook

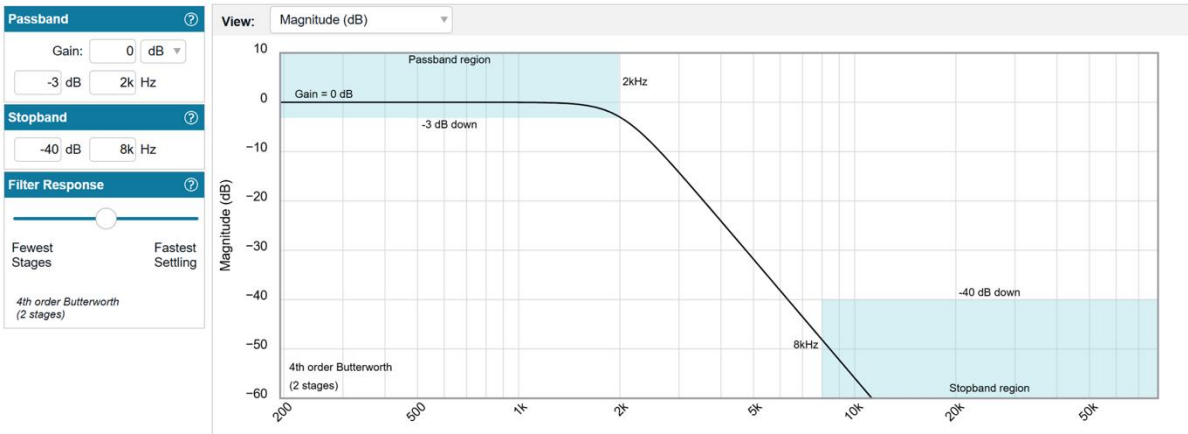
Part 2 | Section 2: Bandwidth and Filtering

Section 2: Bandwidth and Filtering

Before the Lab

The LT1366 operational amplifier has a gain-bandwidth product (GBW) of 400 kHz. Since the LT1366 is not available in the simulation environment, an amplifier with an equivalent GBW was used as a substitute in LTspice.

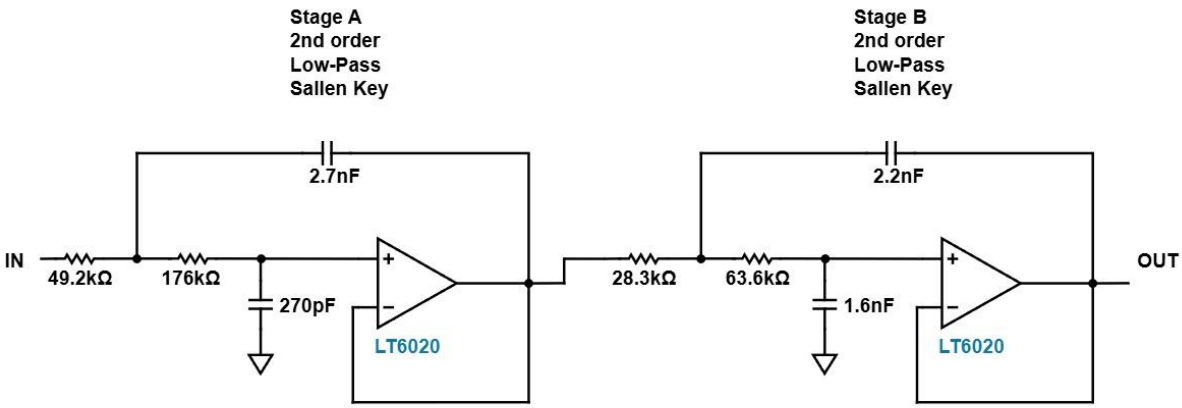
Using the Analog Devices Filter Wizard with the specified passband (0 dB, 2 kHz) and stopband (−40 dB, 8 kHz) constraints, the tool generated the required filter regions and identified a suitable 4th-order Butterworth low-pass topology implemented as two cascaded 2nd-order Sallen-Key stages:



Stage A (2nd-order Low-Pass Sallen-Key): 49.2 kΩ, 170 kΩ, 2.7 nF, 270 pF

Stage B (2nd-order Low-Pass Sallen-Key): 28.3 kΩ, 63.6 kΩ, 2.2 nF, 1.6 nF

Both stages use LT6020 op-amps sourced from the filter design tool.



Gain-Bandwidth Product

To characterise the gain-bandwidth product of the LT1366, a non-inverting amplifier was constructed with a feedback resistor Rf = 68 kΩ and an input resistor R_i = 1.5 kΩ. The supply voltage is ±5 V.

The low-frequency closed-loop gain is:

$$A = Rf / R_i = 68k / 1.5k = 45.3$$

The RC integrator at the input introduces a pole. The transfer function from V_i to V_{out} via the potential divider formed by R and C gives:

$$V_{out} / V_i = Rf / (R + 1/j\omega C) = j\omega CRf / (j\omega RC + 1)$$

The −3 dB frequency of this RC network is:

$$\omega = 1/(RC) = 666.7 \text{ rad/s} \implies f = 666.7 / (2\pi) \approx 106 \text{ Hz}$$

The gain and phase of the amplifier were measured from 100 Hz to 50 kHz:

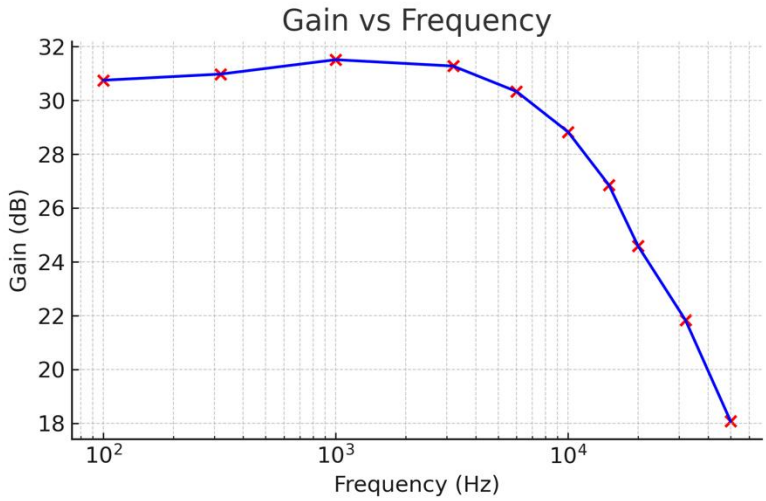
All results are based on experimentally measured data, not on numerical or circuit simulations.

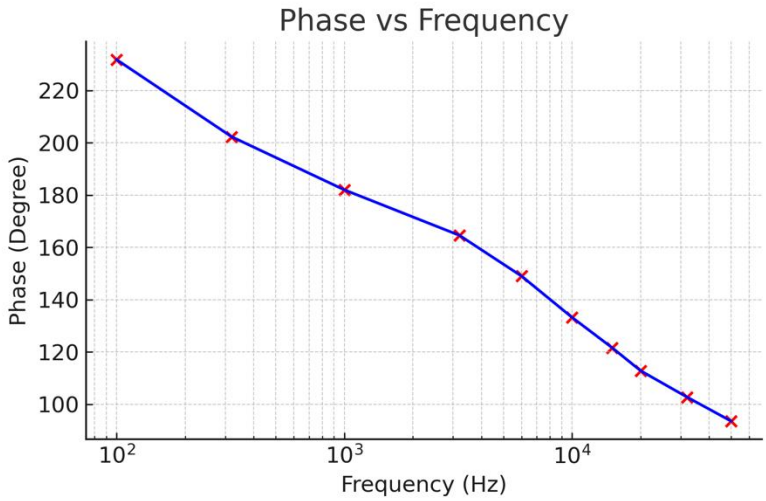
Frequency (Hz)	Vin (mV)	Vout (V)	Gain (dB)	Phase (°)
100	76	2.62	34.47	231.8
320	71.2	2.52	35.39	202.3
1000	69.6	2.62	37.64	182
3200	68.8	2.52	36.63	164.6
6000	68.8	2.26	32.85	149
10000	68	1.9	27.62	133.2
15000	66.4	1.46	21.99	121.6
20000	69.6	1.18	16.95	112.8
32000	61.6	760 mV	12.34	102.7
50000	62.4	500 mV	8.01	93.5

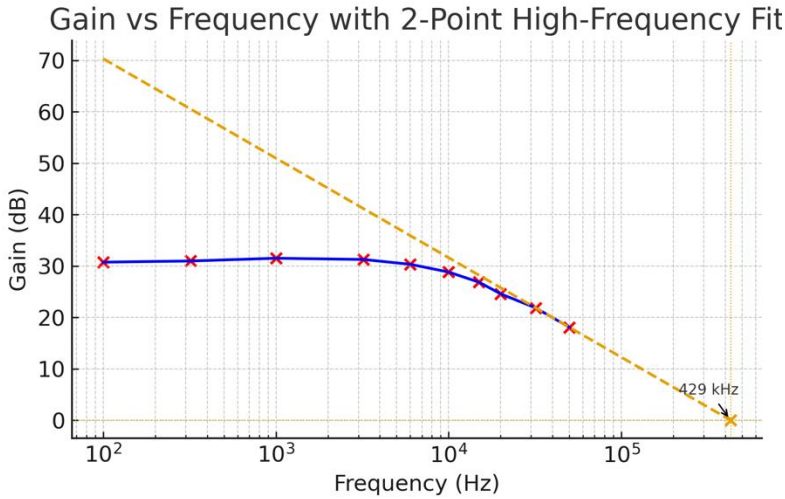
All results are based on experimentally measured data, not on numerical or circuit simulations.

The gain is approximately constant at low frequencies (~31 dB), then rolls off at higher frequencies as expected from the op-amp's finite GBW. A linear extrapolation using the last two data points (32 kHz and 50 kHz) was performed to estimate the frequency at which the gain falls to 0 dB. The x-axis intercept was found at approximately **429 kHz**, which is in close agreement with the specified GBW of 400 kHz, confirming the amplifier's frequency characteristics.

The graphs are in the next page.


Gain vs Frequency
Gain (dB)
Frequency (Hz)


Phase vs Frequency
Phase (Degree)
Frequency (Hz)


Gain vs Frequency with 2-Point High-Frequency Fit
Gain (dB)
429 kHz
Frequency (Hz)

Section 2 Challenge: Active Low-Pass Filter

Circuit Implementation

Based on the 4th-order Butterworth low-pass filter designed in the pre-lab simulation (two cascaded Sallen-Key stages using LT6020), the circuit was assembled on a breadboard using components with values as close as possible to the simulated design. Exact values could not always be sourced from the lab component drawers, and the nearest available substitutes were used.

Experimental Observations

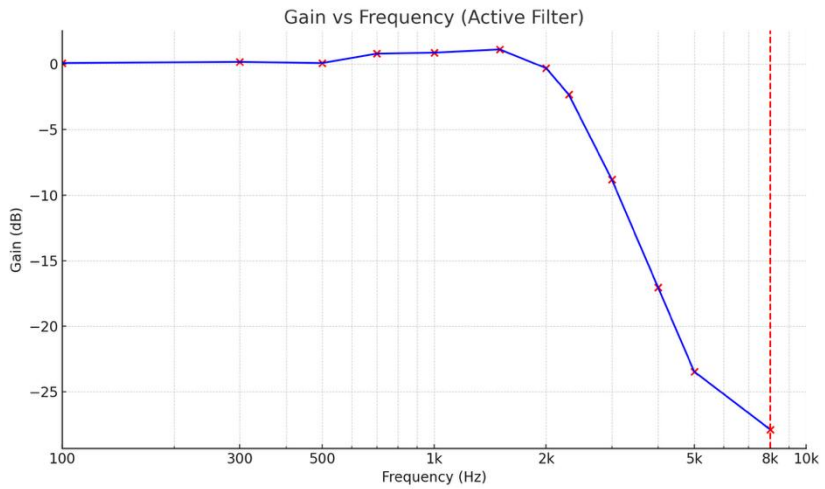
The signal source was set to Mode 8, which outputs a melody superimposed with an 8 kHz sinusoidal whine. If the filter is correctly implemented, the 8 kHz component should be attenuated below the audible threshold in the output, leaving only the melody.

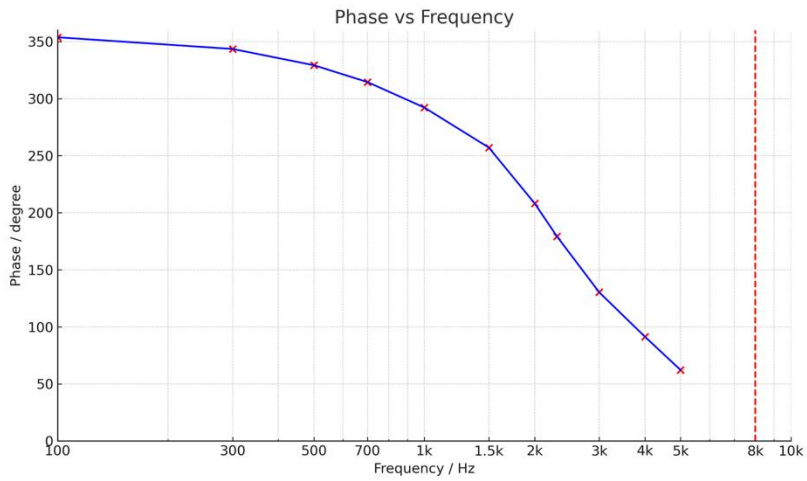
The filter response was characterised over the range 100 Hz to 8 kHz:

Frequency (Hz)	Vin (mV)	Vout (V)	Gain	Phase (°)
100	210	212	1.010	-6.21 / 353.79
300	210	214	1.019	-16.42 / 343.58
500	216	218	1.009	-30.69 / 329.31
700	210	230	1.095	-45.44 / 314.56
1000	210	232	1.105	-67.75 / 292.25
1500	199.5	226.7	1.136	-102.86 / 257.14
2000	199	191.9	0.964	-151.75 / 208.25
2300	200	152.9	0.765	179.34
3000	200.6	72.9	0.363	130.32
4000	198.8	27.9	0.140	91.3
5000	199.7	13.4	0.067	62.21
8000	201.8	8.15	0.040	unmeasurable

All results are based on experimentally measured data, not on numerical or circuit simulations.

At 8 kHz the output amplitude was approximately 8 mV, corresponding to a gain well below 0 dB. However, the attenuation did not reach the theoretical target of −40 dB. Additionally, the output waveform at 8 kHz degenerated to a near-flat line with residual noise, making phase measurement impossible.



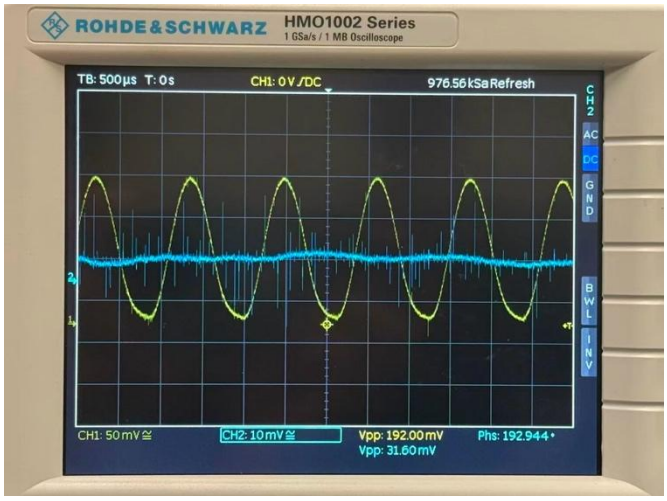


The discrepancy between measured and theoretical attenuation is attributed to two principal causes:

- **Parasitic resistance in capacitors:** Real capacitors include a series equivalent resistance (ESR), which degrades the ideal reactive behaviour assumed in the simulation, particularly at higher frequencies.
- **Component value substitution:** Several resistor and capacitor values specified by the filter design tool were unavailable in the lab. Substituted values shifted the actual pole locations and modified the filter's frequency response relative to the ideal design.

Demonstration

With the active filter connected in-line with the signal source (Mode 8) and an earphone at the output, the 8 kHz whine was significantly reduced, and only the melody remained clearly audible. This confirmed that the filter was operating correctly as a low-pass circuit, even if the stopband attenuation fell short of the −40 dB specification.



Advantages of an Active Filter over a Passive Filter

- **Gain control:** An active filter can provide gain in the passband in addition to frequency-selective attenuation. For this task, a specific stopband attenuation level of −40 dB was required, which can be achieved and adjusted through the op-amp feedback network.
- **High input impedance:** The op-amp input presents a very high impedance to the preceding circuit stage, drawing negligible current. This isolates the filter from the source, preventing

loading effects that would otherwise alter the component values' effective behaviour and shift the filter's characteristic frequencies.